Studi Kasus Penggunaa R untuk Analisis Rancangan Percobaan Pertanian

Nurfian Qodar

Memuat *library*

Memuat *library* dapat menggunakan fungsi bawaan R yaitu `library` atau gunakan pacman `pacman::p_load` (lebih cepat). Gunakan salah satu kode dibawah ini:

- Dengan `pacman` (lebih cepat)

```
if (!("pacman" %in% installed.packages())) {
  install.packages("pacman")
}

pacman::p_load(tidyverse, agricolae, openxlsx, emmeans, nlme)
```

- Dengan `library` R

```
library(tidyverse, agricolae, openxlsx)
```

1. Rancangan acak lengkap (RAL)

1.1. Pratinjau data

a. Muat data

```
df <- read.xlsx("data.xlsx", sheet = "ral_jagung")
```

b. Ringkasan data

- Memeriksa tipe data

```
df |> str()
```

```
'data.frame': 48 obs. of  4 variables:
 $ f1            : chr  "A1" "A1" "A1" "A1" ...
 $ f2            : chr  "B1" "B1" "B1" "B1" ...
 $ rep           : chr  "1" "2" "3" "4" ...
 $ tinggi_tanaman: num  142 145 146 147 147 ...
```

- Ringkasan data

```
df |> summary()
```

```
       f1             f2            rep     tinggi_tanaman
 Length   :48   Length   :48   Length   :48   Min.   :140.7
 N.unique : 4   N.unique : 3   N.unique : 4   1st Qu.:148.4
 N.blank  : 0   N.blank  : 0   N.blank  : 0   Median :156.6
 Min.nchar: 2   Min.nchar: 2   Min.nchar: 1   Mean   :159.5
 Max.nchar: 2   Max.nchar: 2   Max.nchar: 1   3rd Qu.:167.6
                                             Max.   :190.0
```

- Memeriksa data kosong

```
df |> is.na() |> colSums()
```

```
           f1            f2           rep tinggi_tanaman
            0             0             0              0
```

c. Membersihkan data

```
df <- df |>
  rename(
    faktor1 = f1,
    faktor2 = f2,
    ulangan = rep
  ) |>
  mutate(
    faktor1 = as.factor(faktor1),
    faktor2 = as.factor(faktor2),
    ulangan = as.factor(ulangan)
  )
```

Tijauan ulang data setelah dirapikan:

```
df |> str()
```

```
'data.frame':   48 obs. of  4 variables:
 $ faktor1      : Factor w/ 4 levels "A1","A2","A3",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ faktor2      : Factor w/ 3 levels "B1","B2","B3": 1 1 1 1 2 2 2 2 3 3 ...
 $ ulangan      : Factor w/ 4 levels "1","2","3","4": 1 2 3 4 1 2 3 4 1 2 ...
 $ tinggi_tanaman: num  142 145 146 147 147 ...
```

```
df |> summary()
```

```
 faktor1 faktor2 ulangan tinggi_tanaman
 A1:12   B1:16   1:12    Min.   :140.7
 A2:12   B2:16   2:12    1st Qu.:148.4
 A3:12   B3:16   3:12    Median :156.6
 A4:12           4:12    Mean   :159.5
                         3rd Qu.:167.6
                         Max.   :190.0
```

1.2. Analisis data

a. Analisis sidik ragam (ANOVA)

```
model <- aov(tinggi_tanaman ~ faktor1 * faktor2, data = df)
```

Lihat hasil analisis:

```
model |> summary()
```

```
                Df Sum Sq Mean Sq F value   Pr(>F)
faktor1          3 3102.8  1034.3  62.071 2.66e-14 ***
faktor2          2 2973.2  1486.6  89.219 1.12e-14 ***
faktor1:faktor2  6  685.5   114.2   6.857 6.53e-05 ***
Residuals       36  599.9    16.7
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b. Uji lanjut

Karena terdapat interaksi maka uji lanjutan akan dilakukan pada seluruh `faktor1` terhadap setiap `faktor2` dan seluruh `faktor2` terhadap setiap `faktor1`.

DMRT

Uji `faktor1` pada setiap `faktor2`

```
faktor2 <- df$faktor2 |>
    unique()

faktor2
```

```
[1] B1 B2 B3
Levels: B1 B2 B3
```

```r
for (f in faktor2) {
  dd <- df |>
    filter(faktor2 == f)
  model <- aov(tinggi_tanaman ~ faktor1, data = dd)
  dmrt <- duncan.test(model, trt = "faktor1")
  tabel <- dmrt$groups |>
      as.data.frame() |>
      rownames_to_column("Faktor 1")

  paste0("Faktor 1 pada ", f, "\n") |> cat(sep = "")

  print(tabel)

  cat("\n\n")
}
```

```
Faktor 1 pada B1
  Faktor 1 tinggi_tanaman groups
1       A3       155.1360      a
2       A4       152.5830     ab
3       A2       147.0582     bc
4       A1       145.0537      c


Faktor 1 pada B2
  Faktor 1 tinggi_tanaman groups
1       A4       169.0896      a
2       A3       167.0081      a
3       A2       152.7377      b
4       A1       148.8083      b


Faktor 1 pada B3
  Faktor 1 tinggi_tanaman groups
1       A4       185.2355      a
2       A3       173.9264      b
3       A2       165.2311      c
4       A1       152.5464      d
```

Uji `faktor2` pada setiap `faktor1`

```
faktor1 <- df$faktor1 |>
    unique()

faktor1
```

```
[1] A1 A2 A3 A4
Levels: A1 A2 A3 A4
```

```
for (f in faktor1) {
  dd <- df |>
    filter(faktor1 == f)

  model <- aov(tinggi_tanaman ~ faktor2, data = dd)

  dmrt <- duncan.test(model, trt = "faktor2")

  tabel <- dmrt$groups |>
      as.data.frame() |>
      rownames_to_column("Faktor 2")

  paste0("Faktor 2 pada ", f, "\n") |> cat(sep = "")

  print(tabel)

  cat("\n\n")
}
```

```
Faktor 2 pada A1
  Faktor 2 tinggi_tanaman groups
1       B3       152.5464      a
2       B2       148.8083     ab
3       B1       145.0537      b


Faktor 2 pada A2
  Faktor 2 tinggi_tanaman groups
1       B3       165.2311      a
2       B2       152.7377      b
3       B1       147.0582      b
```

```
Faktor 2 pada A3
  Faktor 2 tinggi_tanaman groups
1       B3       173.9264      a
2       B2       167.0081      b
3       B1       155.1360      c


Faktor 2 pada A4
  Faktor 2 tinggi_tanaman groups
1       B3       185.2355      a
2       B2       169.0896      b
3       B1       152.5830      c
```

Tukey HSD

Uji `faktor1` pada setiap `faktor2`

```
for (f in faktor2) {
  dd <- df |>
    filter(faktor2 == f)
  model <- aov(tinggi_tanaman ~ faktor1, data = dd)
  hsd <- TukeyHSD(model, "faktor1")
  tabel <- hsd$faktor1 |>
      as.data.frame() |>
      rownames_to_column("Faktor 1") |>
      mutate(
        signif = `p adj` < 0.05
      )


  paste0("Faktor 1 pada ", f, "\n") |> cat(sep = "")

  print(tabel)

  cat("\n\n")
}
```

```
Faktor 1 pada B1
  Faktor 1      diff         lwr       upr      p adj signif
1    A2-A1  2.004471  -7.0882142 11.097155 0.91195436  FALSE
2    A3-A1 10.082338   0.9896532 19.175023 0.02849686   TRUE
3    A4-A1  7.529293  -1.5633922 16.621978 0.11822712  FALSE
```

```
4    A3-A2  8.077867  -1.0148174 17.170552 0.08791312  FALSE
5    A4-A2  5.524822  -3.5678628 14.617507 0.31812878  FALSE
6    A4-A3 -2.553045 -11.6457302  6.539639 0.83759401  FALSE


Faktor 1 pada B2
  Faktor 1      diff       lwr      upr        p adj signif
1    A2-A1  3.929391 -4.427477 12.28626 0.5249115174  FALSE
2    A3-A1 18.199832  9.842964 26.55670 0.0001570759   TRUE
3    A4-A1 20.281287 11.924419 28.63816 0.0000554481   TRUE
4    A3-A2 14.270441  5.913573 22.62731 0.0013556618   TRUE
5    A4-A2 16.351896  7.995028 24.70876 0.0004195716   TRUE
6    A4-A3  2.081455 -6.275414 10.43832 0.8793345591  FALSE


Faktor 1 pada B3
  Faktor 1      diff       lwr      upr        p adj signif
1    A2-A1 12.684684  4.451195 20.91817 3.088672e-03   TRUE
2    A3-A1 21.379938 13.146449 29.61343 2.831708e-05   TRUE
3    A4-A1 32.689097 24.455608 40.92259 3.114643e-07   TRUE
4    A3-A2  8.695254  0.461766 16.92874 3.745880e-02   TRUE
5    A4-A2 20.004413 11.770925 28.23790 5.483710e-05   TRUE
6    A4-A3 11.309159  3.075670 19.54265 7.216984e-03   TRUE
```

Uji `faktor2` pada setiap `faktor1`

```r
for (f in faktor1) {
  dd <- df |>
    filter(faktor1 == f)

  model <- aov(tinggi_tanaman ~ faktor2, data = dd)

  hsd <- TukeyHSD(model, "faktor2")

  tabel <- hsd$faktor2 |>
      as.data.frame() |>
      rownames_to_column("Faktor 2")

  paste0("Faktor 2 pada ", f, "\n") |> cat(sep = "")

  print(tabel)
```

```
  cat("\n\n")
}
```

```
Faktor 2 pada A1
  Faktor 2     diff       lwr       upr      p adj
1    B2-B1 3.754593 -1.874895  9.384080 0.20491486
2    B3-B1 7.492725  1.863237 13.122212 0.01201387
3    B3-B2 3.738132 -1.891356  9.367619 0.20731760


Faktor 2 pada A2
  Faktor 2      diff       lwr      upr       p adj
1    B2-B1  5.679513 -4.149244 15.50827 0.289316210
2    B3-B1 18.172938  8.344180 28.00170 0.001538702
3    B3-B2 12.493424  2.664667 22.32218 0.015481858


Faktor 2 pada A3
  Faktor 2      diff       lwr      upr        p adj
1    B2-B1 11.872087  3.581725 20.16245 0.0078821030
2    B3-B1 18.790325 10.499962 27.08069 0.0003585448
3    B3-B2  6.918237 -1.372125 15.20860 0.1022722349


Faktor 2 pada A4
  Faktor 2     diff       lwr      upr        p adj
1    B2-B1 16.50659  8.584921 24.42825 6.639053e-04
2    B3-B1 32.65253 24.730863 40.57419 2.945925e-06
3    B3-B2 16.14594  8.224276 24.06761 7.779963e-04
```

2. Rancangan Acak Kelompok (RAK)

2.1. Pratinjau data

a. Muat data

```
df <- read.xlsx("data.xlsx", sheet = "rak_jagung")
```

b. Ringkasan data

- Memeriksa tipe data

```
df |> str()
```

```
'data.frame': 48 obs. of  4 variables:
 $ f1            : chr  "A1" "A1" "A1" "A1" ...
 $ f2            : chr  "B1" "B1" "B1" "B1" ...
 $ rep           : chr  "1" "2" "3" "4" ...
 $ tinggi_tanaman: num  128 141 156 168 135 ...
```

- Ringkasan data

```
df |> summary()
```

```
       f1             f2            rep     tinggi_tanaman
 Length   :48   Length   :48   Length   :48   Min.   :128.0
 N.unique : 4   N.unique : 3   N.unique : 4   1st Qu.:143.0
 N.blank  : 0   N.blank  : 0   N.blank  : 0   Median :161.3
 Min.nchar: 2   Min.nchar: 2   Min.nchar: 1   Mean   :159.4
 Max.nchar: 2   Max.nchar: 2   Max.nchar: 1   3rd Qu.:171.4
                                             Max.   :190.0
```

- Memeriksa data kosong

```
df |> is.na() |> colSums()
```

```
            f1             f2            rep tinggi_tanaman
             0              0              0              0
```

c. Membersihkan data

```
df <- df |>
  rename(
    faktor1 = f1,
    faktor2 = f2,
    ulangan = rep
  ) |>
  mutate(
    faktor1 = as.factor(faktor1),
    faktor2 = as.factor(faktor2),
    ulangan = as.factor(ulangan)
  )
```

Tijauan ulang data setelah dirapikan:

```
df |> str()
```

```
'data.frame':   48 obs. of  4 variables:
 $ faktor1      : Factor w/ 4 levels "A1","A2","A3",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ faktor2      : Factor w/ 3 levels "B1","B2","B3": 1 1 1 1 2 2 2 2 3 3 ...
 $ ulangan      : Factor w/ 4 levels "1","2","3","4": 1 2 3 4 1 2 3 4 1 2 ...
 $ tinggi_tanaman: num  128 141 156 168 135 ...
```

```
df |> summary()
```

```
 faktor1 faktor2 ulangan tinggi_tanaman
 A1:12   B1:16   1:12    Min.   :128.0
 A2:12   B2:16   2:12    1st Qu.:143.0
 A3:12   B3:16   3:12    Median :161.3
 A4:12           4:12    Mean   :159.4
                         3rd Qu.:171.4
                         Max.   :190.0
```

2.2. Analisis data

a. Analisis sidik ragam (ANOVA)

```
model <- aov(tinggi_tanaman ~ faktor1 * faktor2 + ulangan, data = df)
```

Lihat hasil analisis:

```
model |> summary()
```

```
                Df Sum Sq Mean Sq F value   Pr(>F)
faktor1          3   2166   721.9  54.707 6.70e-13 ***
faktor2          2   1405   702.5  53.239 4.69e-11 ***
ulangan          3   8818  2939.4 222.769  < 2e-16 ***
faktor1:faktor2  6    208    34.6   2.622   0.0344 *
Residuals       33    435    13.2
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b. Uji lanjut

Karena terdapat interaksi maka uji lanjutan akan dilakukan pada seluruh `faktor1` terhadap setiap `faktor2` dan seluruh `faktor2` terhadap setiap `faktor1`.

DMRT

Uji `faktor1` pada setiap `faktor2`

```
faktor2 <- df$faktor2 |>
    unique()

faktor2
```

```
[1] B1 B2 B3
Levels: B1 B2 B3
```

```
for (f in faktor2) {
  dd <- df |>
    filter(faktor2 == f)
  model <- aov(tinggi_tanaman ~ faktor1 + ulangan, data = dd)
  dmrt <- duncan.test(model, trt = "faktor1")
  tabel <- dmrt$groups |>
      as.data.frame() |>
      rownames_to_column("Faktor 1")

  paste0("Faktor 1 pada ", f, "\n") |> cat(sep = "")

  print(tabel)

  cat("\n\n")
}
```

```
Faktor 1 pada B1
  Faktor 1 tinggi_tanaman groups
1       A4       159.4615      a
2       A3       154.0748      b
3       A2       148.8423      c
4       A1       147.9089      c


Faktor 1 pada B2
  Faktor 1 tinggi_tanaman groups
1       A4       168.1962      a
2       A3       163.9664     ab
3       A2       157.9523      b
4       A1       148.8018      c
```

```
Faktor 1 pada B3
  Faktor 1 tinggi_tanaman groups
1       A4       175.3367      a
2       A3       172.6735      a
3       A2       161.0408      b
4       A1       154.1871      c
```

Uji `faktor2` pada setiap `faktor1`

```
faktor1 <- df$faktor1 |>
    unique()

faktor1
```

```
[1] A1 A2 A3 A4
Levels: A1 A2 A3 A4
```

```
for (f in faktor1) {
  dd <- df |>
    filter(faktor1 == f)

  model <- aov(tinggi_tanaman ~ faktor2 + ulangan, data = dd)

  dmrt <- duncan.test(model, trt = "faktor2")

  tabel <- dmrt$groups |>
      as.data.frame() |>
      rownames_to_column("Faktor 2")

  paste0("Faktor 2 pada ", f, "\n") |> cat(sep = "")

  print(tabel)

  cat("\n\n")
}
```

```
Faktor 2 pada A1
  Faktor 2 tinggi_tanaman groups
1       B3       154.1871      a
2       B2       148.8018      a
```

```
3       B1       147.9089      a


Faktor 2 pada A2
  Faktor 2 tinggi_tanaman groups
1       B3       161.0408      a
2       B2       157.9523      a
3       B1       148.8423      b


Faktor 2 pada A3
  Faktor 2 tinggi_tanaman groups
1       B3       172.6735      a
2       B2       163.9664      b
3       B1       154.0748      c


Faktor 2 pada A4
  Faktor 2 tinggi_tanaman groups
1       B3       175.3367      a
2       B2       168.1962      b
3       B1       159.4615      c
```

Tukey HSD

Uji `faktor1` pada setiap `faktor2`

```
for (f in faktor2) {
  dd <- df |>
    filter(faktor2 == f)
  model <- aov(tinggi_tanaman ~ faktor1 + ulangan, data = dd)
  hsd <- TukeyHSD(model, "faktor1")
  tabel <- hsd$faktor1 |>
      as.data.frame() |>
      rownames_to_column("Faktor 1") |>
      mutate(
        signif = `p adj` < 0.05
      )


  paste0("Faktor 1 pada ", f, "\n") |> cat(sep = "")

  print(tabel)
```

```
  cat("\n\n")
}
```

```
Faktor 1 pada B1
  Faktor 1       diff          lwr       upr       p adj signif
1    A2-A1  0.9333539 -5.240849918  7.107558 0.963409572  FALSE
2    A3-A1  6.1659021 -0.008301721 12.340106 0.050316900  FALSE
3    A4-A1 11.5525195  5.378315713 17.726723 0.001146769   TRUE
4    A3-A2  5.2325482 -0.941655603 11.406752 0.102077489  FALSE
5    A4-A2 10.6191656  4.444961832 16.793369 0.002074898   TRUE
6    A4-A3  5.3866174 -0.787586366 11.560821 0.090905234  FALSE


Faktor 1 pada B2
  Faktor 1      diff        lwr      upr        p adj signif
1    A2-A1  9.150484  0.2171185 18.08385 0.0446051248   TRUE
2    A3-A1 15.164568  6.2312025 24.09793 0.0022709291   TRUE
3    A4-A1 19.394431 10.4610660 28.32780 0.0003832344   TRUE
4    A3-A2  6.014084 -2.9192815 14.94745 0.2234758592  FALSE
5    A4-A2 10.243947  1.3105820 19.17731 0.0251718707   TRUE
6    A4-A3  4.229863 -4.7035020 13.16323 0.4874018988  FALSE


Faktor 1 pada B3
  Faktor 1      diff        lwr      upr        p adj signif
1    A2-A1  6.853752 -0.8438543 14.55136 8.361256e-02  FALSE
2    A3-A1 18.486418 10.7888110 26.18402 1.764907e-04   TRUE
3    A4-A1 21.149639 13.4520323 28.84725 6.073913e-05   TRUE
4    A3-A2 11.632665  3.9350587 19.33027 4.928331e-03   TRUE
5    A4-A2 14.295887  6.5982799 21.99349 1.209660e-03   TRUE
6    A4-A3  2.663221 -5.0343853 10.36083 7.094984e-01  FALSE
```

Uji `faktor2` pada setiap `faktor1`

```
for (f in faktor1) {
  dd <- df |>
    filter(faktor1 == f)

  model <- aov(tinggi_tanaman ~ faktor2 + ulangan, data = dd)

  hsd <- TukeyHSD(model, "faktor2")

  tabel <- hsd$faktor2 |>
```

```
      as.data.frame() |>
      rownames_to_column("Faktor 2")

  paste0("Faktor 2 pada ", f, "\n") |> cat(sep = "")

  print(tabel)

  cat("\n\n")
}
```

```
Faktor 2 pada A1
  Faktor 2      diff       lwr      upr     p adj
1    B2-B1 0.8928683 -6.789933  8.57567 0.9330726
2    B3-B1 6.2781584 -1.404643 13.96096 0.1011329
3    B3-B2 5.3852901 -2.297511 13.06809 0.1593865


Faktor 2 pada A2
  Faktor 2      diff       lwr      upr       p adj
1    B2-B1  9.109998  1.868321 16.35168 0.019589800
2    B3-B1 12.198557  4.956879 19.44023 0.004989907
3    B3-B2  3.088558 -4.153119 10.33024 0.441189380


Faktor 2 pada A3
  Faktor 2      diff         lwr      upr       p adj
1    B2-B1  9.891534 -0.03883126 19.82190 0.050744479
2    B3-B1 18.598674  8.66830842 28.52904 0.002920539
3    B3-B2  8.707140 -1.22322585 18.63751 0.080165746


Faktor 2 pada A4
  Faktor 2      diff       lwr      upr        p adj
1    B2-B1  8.734780  3.911098 13.55846 0.0034695690
2    B3-B1 15.875278 11.051596 20.69896 0.0001350079
3    B3-B2  7.140497  2.316816 11.96418 0.0093372741
```

3. Rancangan Bujur Sangkar Latin (RBSL)

3.1. Pratinjau data

a. Muat data

```
df <- read.xlsx("data.xlsx", sheet = "rbsl_jagung")
```

b. Ringkasan data

- Memeriksa tipe data

```
df |> str()
```

```
'data.frame': 81 obs. of  8 variables:
 $ baris : num  1 2 3 4 5 6 7 8 9 1 ...
 $ kolom : num  1 1 1 1 1 1 1 1 1 2 ...
 $ trt   : num  1 2 3 4 5 6 7 8 9 2 ...
 $ f1    : chr  "A1" "A2" "A3" "A1" ...
 $ f2    : chr  "B1" "B1" "B1" "B2" ...
 $ f1_num: num  1 2 3 1 2 3 1 2 3 2 ...
 $ f2_num: num  1 1 1 2 2 2 3 3 3 1 ...
 $ data  : num  123 130 153 132 147 ...
```

- Ringkasan data

```
df |> summary()
```

```
     baris       kolom        trt            f1             f2         f1_num
 Min.   :1   Min.   :1   Min.   :1   Length   :81   Length   :81   Min.   :1
 1st Qu.:3   1st Qu.:3   1st Qu.:3   N.unique : 3   N.unique : 3   1st Qu.:1
 Median :5   Median :5   Median :5   N.blank  : 0   N.blank  : 0   Median :2
 Mean   :5   Mean   :5   Mean   :5   Min.nchar: 2   Min.nchar: 2   Mean   :2
 3rd Qu.:7   3rd Qu.:7   3rd Qu.:7   Max.nchar: 2   Max.nchar: 2   3rd Qu.:3
 Max.   :9   Max.   :9   Max.   :9                                 Max.   :3
     f2_num       data
 Min.   :1   Min.   :123.3
 1st Qu.:1   1st Qu.:147.0
 Median :2   Median :155.0
 Mean   :2   Mean   :157.5
 3rd Qu.:3   3rd Qu.:167.0
 Max.   :3   Max.   :190.0
```

- Memeriksa data kosong

```
df |> is.na() |> colSums()
```

```
  baris  kolom    trt     f1     f2 f1_num f2_num   data
     0      0      0      0      0      0      0      0
```

c. Membersihkan data

```
df <- df |>
    rename(
      faktor1 = f1,
      faktor2 = f2
    ) |>
    mutate(
      faktor1 = as.factor(faktor1),
      faktor2 = as.factor(faktor2),
      kolom = as.factor(kolom),
      baris = as.factor(baris)
    ) |>
    select(-trt)
```

Tinjauan ulang data setelah dirapikan:

```
df |> str()
```

```
'data.frame':   81 obs. of  7 variables:
 $ baris  : Factor w/ 9 levels "1","2","3","4",..: 1 2 3 4 5 6 7 8 9 1 ...
 $ kolom  : Factor w/ 9 levels "1","2","3","4",..: 1 1 1 1 1 1 1 1 1 2 ...
 $ faktor1: Factor w/ 3 levels "A1","A2","A3": 1 2 3 1 2 3 1 2 3 2 ...
 $ faktor2: Factor w/ 3 levels "B1","B2","B3": 1 1 1 2 2 2 3 3 3 1 ...
 $ f1_num : num  1 2 3 1 2 3 1 2 3 2 ...
 $ f2_num : num  1 1 1 2 2 2 3 3 3 1 ...
 $ data   : num  123 130 153 132 147 ...
```

```
df |> summary()
```

```
     baris        kolom    faktor1 faktor2     f1_num      f2_num
 1      : 9   1      : 9   A1:27   B1:27   Min.   :1   Min.   :1
 2      : 9   2      : 9   A2:27   B2:27   1st Qu.:1   1st Qu.:1
 3      : 9   3      : 9   A3:27   B3:27   Median :2   Median :2
 4      : 9   4      : 9                   Mean   :2   Mean   :2
 5      : 9   5      : 9                   3rd Qu.:3   3rd Qu.:3
```

```
 6      : 9   6      : 9                  Max.   :3   Max.   :3
 (Other):27   (Other):27
      data
 Min.   :123.3
 1st Qu.:147.0
 Median :155.0
 Mean   :157.5
 3rd Qu.:167.0
 Max.   :190.0
```

3.2. Analisis data

a. Analisis sidik ragam (ANOVA)

```
model <- lm(data ~ kolom + baris + faktor1 * faktor2, data = df)
```

```
model |> anova()
```

```
Analysis of Variance Table

Response: data
                Df Sum Sq Mean Sq F value    Pr(>F)
kolom            8 2587.3  323.41  7.0088 2.288e-06 ***
baris            8 1575.7  196.96  4.2684 0.0004718 ***
faktor1          2 6212.7 3106.36 67.3195 1.269e-15 ***
faktor2          2 4901.2 2450.60 53.1081 1.166e-13 ***
faktor1:faktor2  4  710.8  177.70  3.8509 0.0078105 **
Residuals       56 2584.0   46.14
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b. Uji lanjut

Tukey HSD

```
hsd_f1 <- emmeans(model, ~ faktor1 | faktor2)

pairs(hsd_f1, adjust = "tukey")
```

```
faktor2 = B1:
 contrast estimate  SE df t.ratio p.value
```

```
 A1 - A2     -5.72 3.2 56  -1.785  0.1839
 A1 - A3    -12.79 3.2 56  -3.995  0.0006
 A2 - A3     -7.08 3.2 56  -2.210  0.0782

faktor2 = B2:
 contrast estimate  SE df t.ratio p.value
 A1 - A2    -12.68 3.2 56  -3.961  0.0006
 A1 - A3    -21.90 3.2 56  -6.839 <0.0001
 A2 - A3     -9.21 3.2 56  -2.878  0.0154

faktor2 = B3:
 contrast estimate  SE df t.ratio p.value
 A1 - A2    -11.64 3.2 56  -3.634  0.0017
 A1 - A3    -29.62 3.2 56  -9.250 <0.0001
 A2 - A3    -17.98 3.2 56  -5.616 <0.0001

Results are averaged over the levels of: kolom, baris
P value adjustment: tukey method for comparing a family of 3 estimates
```

```
hsd_f2 <- emmeans(model, ~ faktor2 | faktor1)

pairs(hsd_f2, adjust = "tukey")
```

```
faktor1 = A1:
 contrast estimate  SE df t.ratio p.value
 B1 - B2     -3.86 3.2 56  -1.204  0.4558
 B1 - B3    -11.47 3.2 56  -3.581  0.0020
 B2 - B3     -7.61 3.2 56  -2.377  0.0537

faktor1 = A2:
 contrast estimate  SE df t.ratio p.value
 B1 - B2    -10.82 3.2 56  -3.380  0.0037
 B1 - B3    -17.39 3.2 56  -5.430 <0.0001
 B2 - B3     -6.56 3.2 56  -2.050  0.1098

faktor1 = A3:
 contrast estimate  SE df t.ratio p.value
 B1 - B2    -12.96 3.2 56  -4.048  0.0005
 B1 - B3    -28.29 3.2 56  -8.836 <0.0001
 B2 - B3    -15.33 3.2 56  -4.788 <0.0001
```

```
Results are averaged over the levels of: kolom, baris
P value adjustment: tukey method for comparing a family of 3 estimates
```

4. Rancangan Petak Terbagi (*Split Plot*)

4.1. Pratinjau data

a. Muat data

```
df <- read.xlsx("data.xlsx", sheet = "sp_jagung")
```

b. Ringkasan data

- Memeriksa tipe data

```
df |> str()
```

```
'data.frame': 48 obs. of  4 variables:
 $ f1            : chr  "A1" "A1" "A1" "A1" ...
 $ f2            : chr  "B1" "B1" "B1" "B1" ...
 $ rep           : chr  "1" "2" "3" "4" ...
 $ tinggi_tanaman: num  121 135 152 154 129 ...
```

- Ringkasan data

```
df |> summary()
```

```
        f1             f2            rep     tinggi_tanaman
 Length   :48   Length   :48   Length   :48   Min.   :120.5
 N.unique : 4   N.unique : 3   N.unique : 4   1st Qu.:139.5
 N.blank  : 0   N.blank  : 0   N.blank  : 0   Median :152.7
 Min.nchar: 2   Min.nchar: 2   Min.nchar: 1   Mean   :152.5
 Max.nchar: 2   Max.nchar: 2   Max.nchar: 1   3rd Qu.:163.9
                                             Max.   :190.0
```

- Memeriksa data kosong

```
df |> is.na() |> colSums()
```

```
            f1             f2            rep tinggi_tanaman
             0              0              0              0
```

c. Membersihkan data

```
df <- df |>
  rename(
    faktor1 = f1,
    faktor2 = f2,
    ulangan = rep
  ) |>
  mutate(
    faktor1 = as.factor(faktor1),
    faktor2 = as.factor(faktor2),
    ulangan = as.factor(ulangan)
  )
```

Tijauan ulang data setelah dirapikan:

```
df |> str()
```

```
'data.frame':   48 obs. of  4 variables:
 $ faktor1      : Factor w/ 4 levels "A1","A2","A3",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ faktor2      : Factor w/ 3 levels "B1","B2","B3": 1 1 1 1 2 2 2 2 3 3 ...
 $ ulangan      : Factor w/ 4 levels "1","2","3","4": 1 2 3 4 1 2 3 4 1 2 ...
 $ tinggi_tanaman: num  121 135 152 154 129 ...
```

```
df |> summary()
```

```
 faktor1 faktor2 ulangan tinggi_tanaman
 A1:12   B1:16   1:12    Min.   :120.5
 A2:12   B2:16   2:12    1st Qu.:139.5
 A3:12   B3:16   3:12    Median :152.7
 A4:12           4:12    Mean   :152.5
                         3rd Qu.:163.9
                         Max.   :190.0
```

4.2. Analisis data

a. Analisis sidik ragam (ANOVA)

```
model  <- aov(
  tinggi_tanaman ~ faktor1 * faktor2 + Error(ulangan / faktor1),
  data = df
)
```

```
model |> summary()
```

```
Error: ulangan
          Df Sum Sq Mean Sq F value Pr(>F)
Residuals  3   7822    2607

Error: ulangan:faktor1
          Df Sum Sq Mean Sq F value   Pr(>F)
faktor1    3   2011   670.3   62.22 2.44e-06 ***
Residuals  9     97    10.8
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Error: Within
                Df Sum Sq Mean Sq F value   Pr(>F)
faktor2          2 1480.0   740.0   67.91 1.32e-10 ***
faktor1:faktor2  6  375.3    62.6    5.74 0.000809 ***
Residuals       24  261.5    10.9
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b. Uji lanjut

Karena terdapat interaksi maka uji lanjutan akan dilakukan pada seluruh `faktor1` terhadap setiap `faktor2` dan seluruh `faktor2` terhadap setiap `faktor1`.

Tukey HSD

```
model <- lme(
  tinggi_tanaman ~ faktor1 * faktor2,
  random = ~1 | ulangan/faktor1,
  data = df
)
```

```
hsd_f1 <- emmeans(
  model,
  ~ faktor1 | faktor2
)

pairs(
  hsd_f1,
```

```
  adjust = "tukey"
)
```

```
faktor2 = B1:
 contrast estimate   SE df t.ratio p.value
 A1 - A2    -5.817 2.33  9  -2.496  0.1273
 A1 - A3    -5.272 2.33  9  -2.262  0.1786
 A1 - A4   -10.143 2.33  9  -4.352  0.0082
 A2 - A3     0.546 2.33  9   0.234  0.9952
 A2 - A4    -4.326 2.33  9  -1.856  0.3102
 A3 - A4    -4.872 2.33  9  -2.090  0.2270

faktor2 = B2:
 contrast estimate   SE df t.ratio p.value
 A1 - A2    -8.229 2.33  9  -3.531  0.0271
 A1 - A3   -13.206 2.33  9  -5.666  0.0014
 A1 - A4   -17.626 2.33  9  -7.563  0.0002
 A2 - A3    -4.978 2.33  9  -2.136  0.2132
 A2 - A4    -9.397 2.33  9  -4.032  0.0130
 A3 - A4    -4.420 2.33  9  -1.896  0.2945

faktor2 = B3:
 contrast estimate   SE df t.ratio p.value
 A1 - A2    -5.321 2.33  9  -2.283  0.1732
 A1 - A3   -14.755 2.33  9  -6.331  0.0006
 A1 - A4   -25.373 2.33  9 -10.887 <0.0001
 A2 - A3    -9.434 2.33  9  -4.048  0.0127
 A2 - A4   -20.052 2.33  9  -8.604 <0.0001
 A3 - A4   -10.618 2.33  9  -4.556  0.0062

Degrees-of-freedom method: containment
P value adjustment: tukey method for comparing a family of 4 estimates
```

```
hsd_f1 <- emmeans(
  model,
  ~ faktor1 | faktor2
)

pairs(
  hsd_f1,
  adjust = "tukey"
)
```

```
faktor2 = B1:
 contrast estimate   SE df t.ratio p.value
 A1 - A2    -5.817 2.33  9  -2.496  0.1273
 A1 - A3    -5.272 2.33  9  -2.262  0.1786
 A1 - A4   -10.143 2.33  9  -4.352  0.0082
 A2 - A3     0.546 2.33  9   0.234  0.9952
 A2 - A4    -4.326 2.33  9  -1.856  0.3102
 A3 - A4    -4.872 2.33  9  -2.090  0.2270

faktor2 = B2:
 contrast estimate   SE df t.ratio p.value
 A1 - A2    -8.229 2.33  9  -3.531  0.0271
 A1 - A3   -13.206 2.33  9  -5.666  0.0014
 A1 - A4   -17.626 2.33  9  -7.563  0.0002
 A2 - A3    -4.978 2.33  9  -2.136  0.2132
 A2 - A4    -9.397 2.33  9  -4.032  0.0130
 A3 - A4    -4.420 2.33  9  -1.896  0.2945

faktor2 = B3:
 contrast estimate   SE df t.ratio p.value
 A1 - A2    -5.321 2.33  9  -2.283  0.1732
 A1 - A3   -14.755 2.33  9  -6.331  0.0006
 A1 - A4   -25.373 2.33  9 -10.887 <0.0001
 A2 - A3    -9.434 2.33  9  -4.048  0.0127
 A2 - A4   -20.052 2.33  9  -8.604 <0.0001
 A3 - A4   -10.618 2.33  9  -4.556  0.0062

Degrees-of-freedom method: containment
P value adjustment: tukey method for comparing a family of 4 estimates
```

```
hsd_f2 <- emmeans(
  model,
  ~ faktor2 | faktor1
)

pairs(
  hsd_f2,
  adjust = "tukey"
)
```

```
faktor1 = A1:
 contrast estimate   SE df t.ratio p.value
```

```
 B1 - B2     -2.46 2.33 24  -1.056  0.5498
 B1 - B3     -7.55 2.33 24  -3.238  0.0095
 B2 - B3     -5.09 2.33 24  -2.182  0.0949

faktor1 = A2:
 contrast estimate   SE df t.ratio p.value
 B1 - B2     -4.87 2.33 24  -2.091  0.1130
 B1 - B3     -7.05 2.33 24  -3.025  0.0155
 B2 - B3     -2.18 2.33 24  -0.934  0.6244

faktor1 = A3:
 contrast estimate   SE df t.ratio p.value
 B1 - B2    -10.40 2.33 24  -4.461  0.0005
 B1 - B3    -17.03 2.33 24  -7.307 <0.0001
 B2 - B3     -6.63 2.33 24  -2.846  0.0234

faktor1 = A4:
 contrast estimate   SE df t.ratio p.value
 B1 - B2     -9.94 2.33 24  -4.267  0.0008
 B1 - B3    -22.78 2.33 24  -9.773 <0.0001
 B2 - B3    -12.83 2.33 24  -5.506 <0.0001

Degrees-of-freedom method: containment
P value adjustment: tukey method for comparing a family of 3 estimates
```